

# HONGYU HÈ

📍 Princeton, NJ, USA    ✉ [hhy@princeton.edu](mailto:hhy@princeton.edu)    ☎ (640)-230-4640  
🐙 [github.com/HongyuHe](https://github.com/HongyuHe)    🌐 [hhy.ee.princeton.edu](http://hhy.ee.princeton.edu)    📺 [@Hongyu\\_He](https://www.youtube.com/@Hongyu_He)  
( Version: September 2026 | <sup>□</sup> indicates a hyperlink )

## EDUCATION

|                                                                                                                  |                       |
|------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Princeton University — <i>PhD in Electrical Engineering</i></b>                                               | Aug. 2024 – present   |
| ▶ Committee: Maria Apostolaki <sup>□</sup> (advisor), Sharad Malik <sup>□</sup> , Aarti Gupta <sup>□</sup>       | United States         |
| ▶ GPA: 3.95/4.0 (8 graduate-level courses in EE and CS)                                                          |                       |
| <b>ETH Zürich — <i>MSc in Computer Science</i></b>                                                               | Aug. 2021 – Jul. 2024 |
| ▶ Direct Doctoral Scholarship <sup>□</sup> (5 out of 230 CS master's students in 2022)                           | Switzerland           |
| ▶ Thesis awarded highest grade: 6.0/6.0 (Advisor: Gustavo Alonso <sup>□</sup> )                                  |                       |
| ▶ Associated Researcher <sup>□</sup> at ETH AI Center <sup>□</sup>                                               |                       |
| <b>Vrije Universiteit Amsterdam — <i>BSc (Hons) in Computer Science</i></b>                                      | Aug. 2018 – Jul. 2021 |
| <b>Universiteit van Amsterdam — <i>BSc Minor in Mathematics</i></b>                                              | Sep. 2019 – Jan. 2021 |
| ▶ GPA rank: 1st/180 <sup>□</sup> ; national top 8 in CS <sup>□</sup> (KHMW <sup>□</sup> 2020)                    | The Netherlands       |
| <b>Tianjin University — <i>BA in Law</i> (unfinished)</b>                                                        | Sep. 2016 – Aug. 2017 |
| ▶ Top 0.2% among 153,276 regional liberal arts students in National College Entrance Exam, “Gaokao” <sup>□</sup> | China                 |

## INDUSTRY EXPERIENCE\*

|                                                                                                                         |                       |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Microsoft<sup>□</sup> — <i>Research Intern</i></b>   Networking Research Group                                       | Jun. 2026 – Sep. 2026 |
| <b>Curieo AI<sup>□</sup> (sold) — <i>ML Research Fellow</i></b>   remote, part-time                                     | Jul. 2024 – Oct. 2024 |
| <b>IBM Research<sup>□</sup> — <i>Engineering Intern</i></b>   Geodata Team   part-time                                  | Apr. 2024 – Jun. 2024 |
| <b>Apple Inc.<sup>□</sup> — <i>AI/ML Intern</i></b>   Visual Intelligence Team   (return offer)                         | May 2023 – Oct. 2023  |
| <b>Oracle Labs<sup>□</sup> — <i>Graal Cloud Native Intern</i></b>   Serverless Team   part-time   (return-intern offer) | Feb. 2023 – May 2023  |
| <b>Dexter Energy Services<sup>□</sup> — <i>BSc Thesis Intern</i></b>                                                    | Jun. 2021 – Sep. 2021 |
| <b>Huawei Technologies<sup>□</sup> (Amsterdam Research Center) — <i>R&amp;D Intern</i></b>   Search Team                | Mar. 2021 – Jun. 2021 |
| <b>Picnic Technologies<sup>□</sup> — <i>Software Developer</i></b>   Store Team   part-time                             | Sep. 2020 – Mar. 2021 |
| — <b><i>Backend Engineer Intern</i></b>   Mentor: Sander Mak <sup>□</sup>   (return offer)                              | Jun. 2020 – Sep. 2020 |

## TEACHING

|                                                                                                         |            |
|---------------------------------------------------------------------------------------------------------|------------|
| <b>Princeton University</b>   20 hrs/wk (head TA)   Computer Networks <sup>□</sup>                      | 2025       |
| <b>ETH Zürich</b>   14 hrs/wk   Distributed Systems Lab, Big Data (my session recordings <sup>□</sup> ) | 2023, 2024 |
| <b>Vrije Universiteit Amsterdam</b>   8–16 hrs/wk   Graph Theory, Linear Algebra, C++, Calculus I       | 2019, 2020 |

## TECHNICAL SKILLS

**Programming & Scripting:** C/C++, Python, (System)Verilog, Golang, Shell, Scala, Java, SQL,  $\LaTeX$   
**Software & Tools:** Kubernetes, Xilinx Vivado/Vitis suite, JAX, PyTorch, Linux, Git, Docker, Knative, Apache Spark, SLURM, STM32 (ARM Cortex-M3), MongoDB, PostgreSQL, Hadoop, HBase, Spring5, Qt5

\*Industry recommendations: [hhy.ee.princeton.edu/rec/](http://hhy.ee.princeton.edu/rec/)<sup>□</sup>

## PUBLICATIONS

### Preprint

- [14] H. Hè, M. Apostolaki. *Let AI Agents Translate Networks, Not Reason About Them*. [PDF<sup>□</sup>]

### Peer-reviewed paper

- [13] H. Hè, M. Jin, M. Apostolaki. *Worst-Case Discovery and Runtime Protection for RL-Based Network Controllers*. USENIX NSDI 2027 (to appear) [PDF<sup>□</sup>]
- [12] H. Hè<sup>\*</sup>, A. Krentsel<sup>\*</sup>, S. Ratnasamy, M. Apostolaki. *Invariant Discovery for Networked Systems*. ACM HotNets 2026 (to appear) [PDF<sup>□</sup>]
- [11] C. Constantine, H. Hè, M. Apostolaki. *No Hyperscaler? No Problem. Your Students also Break (and Fix) Networks*. ACM HotNets 2026 (to appear)
- [10] M. Jin, H. Hè, M. Apostolaki. *Cross-Flow Correlations Survive Synthesis: Measuring Source-Level Privacy Leakage in Synthetic Network Traces*. NDSS 2027 (to appear) [PDF<sup>□</sup>]
- [9] H. Hè, M. Jin, M. Apostolaki. *Making Logic a First-Class Citizen in Generative ML for Networking*. USENIX NSDI 2026 [code<sup>□</sup>] [PDF<sup>□</sup>]
- [8] H. Hè, M. Apostolaki. *Just-in-Time Logic Enforcement: A new paradigm of combining statistical and symbolic reasoning for network management*. ACM HotNets 2025 [code<sup>□</sup>] [PDF<sup>□</sup>]
- [7] A. Zhou, C. Costic, H. Hè, A. Ghalayini, A. Kabbani, M. Apostolaki. *Mitigating Inter-datacenter Incast with a Proxy: The shortest path is not necessarily the fastest*. ACM HotNets 2025 [PDF<sup>□</sup>]
- [6] B. McKinzie, Z. Gan, J. Fauconnier, ..., H. Hè, ..., P. Gräsch, A. Toshev, Y. Yang. *MMI: Methods, Analysis & Insights from Multimodal LLM Pre-training*. ECCV 2024 **(first flagship multimodal LLM from 🍏)** [PDF<sup>□</sup>]
- [5] H. Hè, M. Friedman, T. Rekatsinas. *EnergAt: Fine-Grained Energy Attribution for Multi-Tenancy*. ACM HotCarbon 2023 **(adopted by Trycarbonara<sup>□</sup>)** [media<sup>□</sup>] [code<sup>□</sup>] [PDF<sup>□</sup>]
- [4] B. Kruit, H. He, J. Urbani. *Tab2Know: Building a Knowledge Base from Tables in Scientific Papers*. ISWC 2020. [code<sup>□</sup>] [PDF<sup>□</sup>]

### Patent

- [3] S. Kakarla, R. Beckett, H. Hè. *Autoformulation of Executable Cost-Saving Rules from Cloud Billing Policies with AI*. Under review.

### Degree thesis

- [2] H. Hè. *FPGA-based SmartNIC for Distributed Machine Learning*. 2024 **(best grade: 6.0/6.0)** [MSc thesis<sup>□</sup>]
- [1] H. He. *How Can Datacenters Join the Smart Grid to Address the Climate Crisis? Using simulation to explore power and cost effects of direct participation in the energy market*. 2021 **(Amsterdam Thesis Award<sup>□</sup>)** [BSc thesis<sup>□</sup>]

## OTHER ACHIEVEMENTS

- ◆ Nominee for Princeton Teaching Award (3 nominees/department annually, see letters from my students here<sup>□</sup>) 2026
- ◆ Princeton Natural & Artificial Mind Fellowship in AI 2025
- ◆ Princeton University Administrative Fellowship in CS 2024
- ◆ Nominee for ETH Medal<sup>□</sup> (MSc) 2023
- ◆ Danish Government scholarship<sup>□</sup> [€ 28,260] 2021
- ◆ Leiden Excellence scholarship<sup>□</sup> [€ 15,000] 2021
- ◆ GRE 334 (verbal: 161, quant: 170, writing: 3) 2021
- ◆ GLOBE scholarship (ranked 1st out of 900+<sup>□</sup>) [€ 1,250] 2020
- ◆ ACM-ICPC: Amsterdam Algorithm Programming Preliminaries (AAPP) with my team, ranked 1st 2020
- ◆ ACM-ICPC: Benelux Algorithm Programming Contest (BAPC) with my team, ranked 9th 2019
- ◆ Young Talent Award<sup>□</sup> by *The Royal Holland Society of Sciences (KHMW)*<sup>□</sup> 2019
- ◆ Ranked 1st in Operating Systems (Cristiano Giuffrida<sup>□</sup>) and Networks (Andrew Tanenbaum<sup>□</sup>) courses 2019

## LANGUAGES

**English:** TOEFL (speaking): 28, IELTS: 9 | **German:** B1 | **Dutch:** Basic | **Mandarin:** PSC<sup>□</sup>: Level 1-B, regional max.